Variational Characterization of the Projection onto a Closed Convex Set

Optimization (IMCA) — complement to ch4, Condiciones de optimalidad

Throughout, $C \subseteq \mathbb{R}^n$ is nonempty, closed and convex, $y \in \mathbb{R}^n$, and $\|\cdot\|$ is the Euclidean norm induced by $\langle \cdot, \cdot \rangle$. The projection $\mathrm{proj}_C(y)$ is the point $p \in C$ satisfying

$$\|y - p\| \le \|y - x\| \qquad \text{for all } x \in C.$$

The following lemma shows this point exists and is unique, so proj_C is well defined.

Lemma 1 (Existence and uniqueness)**.** *There exists exactly one $p \in C$ with $\|y-p\| = \inf_{x\in C} \|y-x\|$.*

Proof. Existence. Fix $x_0 \in C$ and set $r = \|y - x_0\|$. Let $K = C \cap \bar{B}(y, r)$, where $\bar{B}(y, r) = \{x : \|x - y\| \le r\}$. The set K is nonempty ($x_0 \in K$), closed (intersection of two closed sets) and bounded, hence compact by the Heine–Borel theorem. The function $x \mapsto \|y - x\|$ is continuous, so by the Weierstrass extreme value theorem it attains its minimum on K at some $p \in K$. If $x \in C \setminus K$, then $\|y - x\| > r = \|y - x_0\| \ge \|y - p\|$. Hence $\|y - p\| \le \|y - x\|$ for all $x \in C$.

Uniqueness. Let $p_1, p_2 \in C$ both attain $d = \inf_{x\in C} \|y - x\|$, and put $m = \frac{1}{2}(p_1 + p_2)$, which lies in C by convexity. With $a = y - p_1$, $b = y - p_2$ we have $a + b = 2(y - m)$ and $a - b = p_2 - p_1$, so the parallelogram identity $\|a + b\|^2 + \|a - b\|^2 = 2\|a\|^2 + 2\|b\|^2$ gives

$$4\|y - m\|^2 = 2d^2 + 2d^2 - \|p_1 - p_2\|^2.$$

Since $m \in C$, $\|y - m\| \ge d$, i.e. $4\|y - m\|^2 \ge 4d^2$. Therefore $\|p_1 - p_2\|^2 \le 0$, and $p_1 = p_2$. □

Theorem 2 (Variational inequality)**.** *Let $p \in C$. Then*

$$p = \mathrm{proj}_C(y) \quad \Longleftrightarrow \quad \langle y - p, x - p \rangle \le 0 \quad \textit{for all } x \in C.$$

Proof. ($\Rightarrow$) Suppose $p = \mathrm{proj}_C(y)$ and fix $x \in C$. For $t \in (0, 1]$ define $x_t = p + t(x - p) = (1 - t)p + tx$, which belongs to C by convexity. By minimality of p,

$$\|y - p\|^2 \le \|y - x_t\|^2 = \|(y - p) - t(x - p)\|^2 = \|y - p\|^2 - 2t\langle y - p, x - p\rangle + t^2\|x - p\|^2.$$

Cancelling $\|y - p\|^2$ and dividing by $2t > 0$,

$$\langle y - p, x - p \rangle \le \frac{t}{2} \|x - p\|^2 \qquad \text{for all } t \in (0, 1].$$

The left-hand side does not depend on t; letting $t \to 0^+$ yields $\langle y - p, x - p \rangle \le 0$. Since $x \in C$ was arbitrary, the inequality holds on C.

($\Leftarrow$) Suppose $\langle y - p, x - p \rangle \le 0$ for all $x \in C$, and fix $x \in C$. Writing $y - x = (y - p) - (x - p)$,

$$\|y - x\|^2 = \|y - p\|^2 - 2\langle y - p, x - p\rangle + \|x - p\|^2 \;\ge\; \|y - p\|^2 + \|x - p\|^2 \;\ge\; \|y - p\|^2.$$

Thus $\|y - p\| \le \|y - x\|$ for every $x \in C$, and since $p \in C$, p is a minimizer of $x \mapsto \|y - x\|$ over C. By Lemma 1, $p = \mathrm{proj}_C(y)$. □

Theorem 3 (Proposition 1.4 of ch4)**.** *Let f be convex and of class C^1 on an open set containing C, and let $\alpha > 0$. For $\hat{x} \in C$,*

$$\hat{x} \text{ minimizes } f \text{ over } C \iff \hat{x} = \mathrm{proj}_C\big(\hat{x} - \alpha\nabla f(\hat{x})\big).$$

Proof. Throughout, set $y = \hat{x} - \alpha\nabla f(\hat{x})$, so that $y - \hat{x} = -\alpha\nabla f(\hat{x})$.

($\Rightarrow$) Suppose $\hat{x}$ minimizes f over C. A global minimizer is a local one, so by Proposition 1.3(i) of ch4,

$$\langle \nabla f(\hat{x}), x - \hat{x}\rangle \geq 0 \qquad \text{for all } x \in C.$$

Multiplying by $-\alpha < 0$ reverses the inequality:

$$\langle -\alpha\nabla f(\hat{x}), x - \hat{x}\rangle \leq 0, \quad \text{that is,} \quad \langle y - \hat{x}, x - \hat{x}\rangle \leq 0 \qquad \text{for all } x \in C.$$

This is the right-hand side of Theorem 2 for the point $p = \hat{x}$, which lies in C. By the implication ($\Leftarrow$) of Theorem 2, $\hat{x} = \mathrm{proj}_C(y)$, i.e.

$$\hat{x} = \mathrm{proj}_C\big(\hat{x} - \alpha\nabla f(\hat{x})\big).$$

($\Leftarrow$) Suppose $\hat{x} = \mathrm{proj}_C\big(\hat{x} - \alpha\nabla f(\hat{x})\big) = \mathrm{proj}_C(y)$. By the implication ($\Rightarrow$) of Theorem 2 with $p = \hat{x}$,

$$\langle y - \hat{x}, x - \hat{x}\rangle \leq 0, \quad \text{that is,} \quad -\alpha\langle \nabla f(\hat{x}), x - \hat{x}\rangle \leq 0 \qquad \text{for all } x \in C.$$

Dividing by $-\alpha < 0$ gives $\langle \nabla f(\hat{x}), x - \hat{x}\rangle \geq 0$ for all $x \in C$. Since f is convex, Proposition 1.3(ii) of ch4 shows that $\hat{x}$ minimizes f over C. □

Remark 4 (Two different roles)**.** In Theorem 3 the point being projected is $y = \hat{x} - \alpha\nabla f(\hat{x})$, a gradient step from $\hat{x}$, and the candidate projection is $p = \hat{x}$. The projection characterization is used only to translate "$\hat{x}$ is the projection of y" into an inequality; the optimality of $\hat{x}$ for f then comes from Proposition 1.3. The fixed-point form is what the projected gradient method $x_{k+1} = \mathrm{proj}_C(x_k - \alpha\nabla f(x_k))$ iterates: its fixed points are exactly the minimizers of f over C.

Remark 5 (Geometry and first-order optimality)**.** The inequality says that for every $x \in C$ the angle between $y - p$ and $x - p$ is at least $\pi/2$: the set C lies in the closed half-space $\{x : \langle y - p, x - p\rangle \leq 0\}$, whose boundary hyperplane passes through p and is orthogonal to $y - p$. The proof of ($\Leftarrow$) also shows strictness: $\|y - x\|^2 \geq \|y - p\|^2 + \|x - p\|^2$, so $\|y - x\| > \|y - p\|$ whenever $x \neq p$.

Example 6 (Projection onto a closed ball)**.** Let $C = \bar{B}(0, \Delta) = \{x : \|x\| \leq \Delta\}$ with $\Delta > 0$, and let $\|y\| > \Delta$. The candidate is the radial projection $p = \Delta u$ with $u = y/\|y\|$; clearly $p \in C$. Then $y - p = (\|y\| - \Delta)\, u$ and, for $x \in C$,

$$\langle y - p, x - p\rangle = (\|y\| - \Delta)\big(\langle u, x\rangle - \Delta\langle u, u\rangle\big) = (\|y\| - \Delta)\big(\langle u, x\rangle - \Delta\big) \leq (\|y\| - \Delta)(\|x\| - \Delta) \leq 0,$$

using the Cauchy–Schwarz inequality $\langle u, x\rangle \leq \|u\|\|x\| = \|x\|$, then $\|x\| \leq \Delta$ and $\|y\| - \Delta > 0$. By Theorem 2, $\mathrm{proj}_C(y) = \Delta\, y/\|y\|$. (If $\|y\| \leq \Delta$, then $p = y$ trivially satisfies the inequality, so $\mathrm{proj}_C(y) = y$.)

This is the radial projection that appears in Figure 1 of ch5 (trust region). There the trust-region step p_k^* differs from the radial projection of p_k^N, because p_k^* minimizes m_k, not the distance to p_k^N, unless B_k is a positive multiple of I.

Example 7 (Convexity is needed for ($\Rightarrow$))**.** Let $C = \{x \in \mathbb{R}^2 : \|x\| = 1\}$ (closed, not convex), $y = (\frac{1}{2}, 0)$, $p = (1, 0)$. For every $x \in C$, $\|y - x\|^2 = \frac{5}{4} - x_1 \geq \frac{1}{4} = \|y - p\|^2$, so p is a nearest point of C to y. Yet for $x = (-1, 0) \in C$,

$$\langle y - p, x - p\rangle = \langle (-\tfrac{1}{2}, 0), (-2, 0)\rangle = 1 > 0.$$

The proof of ($\Rightarrow$) breaks exactly where $x_t \in C$ was used. Note that ($\Leftarrow$) never used convexity or closedness: any $p \in C$ satisfying the inequality is the unique nearest point of C to y.

Example 8 (Closedness is needed for existence)**.** Let $C = \{x \in \mathbb{R}^2 : \|x\| < 1\}$ (convex, not closed) and $y = (2, 0)$. Then $\inf_{x \in C} \|y - x\| = 1$, but $\|y - x\| \geq 2 - \|x\| > 1$ for all $x \in C$, so no nearest point exists and $\mathrm{proj}_C(y)$ is undefined; correspondingly, no $p \in C$ satisfies the inequality.